UK Train Rides Project

Team Plan for 6 Students: roles, responsibilities and handoffs

1. The project in one sentence

You are a data analyst for a UK railway company. You take raw ticket-sales data, clean it, find insights, predict the future, and present everything in a Tableau dashboard that managers can use to make decisions.

2. The key idea: who waits for whom?

- **One bottleneck:** the cleaned dataset (Student 1) unlocks almost everyone else.
- **Nobody sits idle:** while waiting, each student prepares using the raw data and learning.
- **Handoff rule:** S1 delivers a "good enough" first cleaned version quickly (v1), then improves it (v2, v3...).

3. The 6 roles

#	Role	Main job
S1	Data Cleaning Lead	Cleans the raw data with pandas and owns the cleaned dataset.
S2	Data Model and SQL Lead	Designs the star schema, builds the SQL database, writes SQL queries.
S3	Feature Engineering + Operations Analyst	Adds new columns, then analyzes delays, cancellations and refunds.
S4	Quality Checker + Revenue Analyst	Validates the cleaning, then analyzes revenue, tickets and customers.
S5	Forecasting Lead	Forecasts daily rides and daily revenue for next month.
S6	Forecasting (Ticket Class) + Report Lead	Forecasts demand per ticket class, then leads the report and presentation.

Tableau dashboard: S3 and S4 build it together.

4. What each student is required to do

Each block below lists the student's tasks in the order they should be done, what they must hand in, and how they know their work is finished.

S1: Data Cleaning Lead

Main job	Clean the raw data with pandas and own the cleaned dataset. Everyone else depends on this work, so speed of the first version matters.
Tools	Python (pandas, Matplotlib), Jupyter notebook
Receives from	S4 (validation report listing problems found in the cleaning).
Hands over to	S2, S3 and S4 (first cleaned draft and data dictionary); everyone (final cleaned dataset).

What you must do

1. Explore the raw data: number of rows and columns, data types, a sample of rows, summary statistics, and the count of missing values per column.
2. Find the problems: missing values, duplicate rows, wrong data types (for example dates stored as text), and inconsistent text (for example the same station written in different ways).
3. Clean each problem and record every cleaning step in the preprocessing notebook so it can be repeated.
4. Deliver a "good enough" first version quickly as **clean_v1.csv**, together with the **data dictionary** (a short list of every column and its meaning).
5. Read S4's validation report, fix the problems it lists, and deliver **clean_v2.csv** (or later versions) to everyone.
6. Reproduce the answers to the analysis questions in pandas, make Matplotlib charts, and help S3 and S4 review their work for mistakes.
7. Prepare clean daily tables for S5 and S6, and help evaluate their forecasting models.
8. Write and present your own section of the final report.

What you deliver

- clean_v1.csv, then clean_v2.csv or later (every version numbered)
- Data dictionary
- Preprocessing notebook (shared with S3, reviewed by S4)
- pandas answers and Matplotlib charts for the analysis questions
- Clean daily tables for the forecasting students

You are done when

S4's checklist passes on your cleaned file and the other students confirm the data is usable.

S2: Data Model and SQL Lead

Main job	Design the data model, build the SQL database, and write the SQL queries that answer the analysis questions.
Tools	SQL, database tool, diagramming tool
Receives from	S1 (clean_v1 and the final cleaned dataset).
Hands over to	Everyone (loaded SQL database); S5, S6 and Tableau (exported tables).

What you must do

1. While S1 cleans, design the data model as a **star schema**: one central **Rides** table connected to the **Stations**, **TicketTypes** and **Dates** tables.
2. Draw the data model diagram showing the tables, their columns and how they connect.
3. Write the **CREATE TABLE** code for every table, including primary keys and links between tables.
4. Load the cleaned data into the database as soon as S1 delivers it, and reload when a newer cleaned version arrives.
5. Write the SQL queries that answer both the operations questions (from S3) and the revenue questions (from S4).
6. Export the final tables that Tableau needs.
7. Double-check the numbers in the final report, then write and present your own section.

What you deliver

- Data model diagram
- CREATE TABLE scripts and a loaded SQL database
- SQL file with all analysis queries, each labeled with the question it answers
- Exported tables for Tableau

You are done when

The database loads without errors, the table connections work, and every query returns results that match the pandas answers from S1.

S3: Feature Engineering + Operations Analyst

Main job	Create the new columns the team needs, then analyze operations: delays, cancellations and refunds.
Tools	Python (pandas, Matplotlib), SQL queries from S2, Tableau
Receives from	S1 (cleaned dataset); S2 (SQL queries and database).
Hands over to	S4, S5 and S6 (feature columns); S6 (Tableau dashboard, built together with S4).

What you must do

1. Prepare the feature code on a small raw sample: day of week, month, hour, journey duration, and delay in minutes.
2. Run the feature code on S1's cleaned data and share the new columns with S4, S5 and S6.
3. Write the operations analysis questions, such as: delays by route and by time, cancellations, refund requests, and reasons for delay.
4. Answer these questions with the help of S2 (SQL) and S1 (pandas), and make a sample chart for each one.
5. Start the Tableau layouts using the analysis answers.
6. Build the Tableau dashboard together with S4, including both the analysis answers and the forecasts.
7. Write and present your own section of the final report.

What you deliver

- Feature columns added to the cleaned data (code in the preprocessing notebook)
- List of operations questions, each marked as answered by SQL or pandas, with a sample chart
- Tableau dashboard (shared with S4)

You are done when

The new columns are correct on the full dataset, every operations question has an answer and a chart, and the dashboard shows them clearly.

S4: Quality Checker + Revenue Analyst

Main job	Check that the cleaning is correct, then analyze revenue, tickets and customers.
Tools	Python (pandas, Matplotlib), SQL queries from S2, Tableau
Receives from	S1 (cleaned dataset); S3 (feature columns); S2 (SQL queries and database).
Hands over to	S1 (validation report); S6 (Tableau dashboard, built together with S3).

What you must do

1. Write a **checklist** of what clean data should look like: no empty values in key columns, prices above 0, arrival time after departure time, and so on.
2. Test S1's cleaned output against the checklist.
3. Write a **validation report** listing every problem found and send it to S1.
4. Re-test each new cleaned version until the checklist passes.
5. Write the revenue analysis questions, such as: best routes, ticket types, ticket classes, railcards, and online versus station purchases.
6. Answer these questions with the help of S2 (SQL) and S1 (pandas), and make a sample chart for each one.
7. Start the Tableau layouts, then build the dashboard together with S3.
8. Write and present your own section of the final report.

What you deliver

- Quality checklist
- Validation report for each cleaned version
- List of revenue questions, each marked as answered by SQL or pandas, with a sample chart
- Tableau dashboard (shared with S3)

You are done when

The cleaned data passes every item on the checklist, every revenue question has an answer and a chart, and the dashboard is complete.

S5: Forecasting Lead

Main job	Predict the number of rides and the revenue for each day of the next month.
Tools	Python (scikit-learn, pandas, Matplotlib)
Receives from	S1 (clean daily tables); S3 (feature columns).
Hands over to	S3, S4 and S6 (forecast plots and numbers).

What you must do

1. Learn time series basics: **trend** (the general direction over time) and **seasonality** (patterns that repeat, such as weekly or yearly cycles).
2. Practice grouping the raw data into daily rides.
3. Build the daily aggregated tables with S1 and train first test models.
4. Forecast the **number of rides for next month**.
5. Forecast **daily revenue** for each day of next month, calculated as rides x average price.
6. Evaluate the models with S1, plot the forecasts against past data, and explain how reliable they are.
7. Write and present your own section of the final report.

What you deliver

- Daily aggregated table of rides and revenue
- Forecast plots for rides and daily revenue
- Forecast numbers for the next month, ready for Tableau

You are done when

The forecasts cover every day of next month, the plots are clear, and the model is tested against past data.

S6: Forecasting (Ticket Class) + Report Lead

Main job	Predict the demand for each ticket class, and lead the final report and presentation.
Tools	GitHub, Python (scikit-learn, pandas, Matplotlib), documents and slides
Receives from	S1 (clean daily tables); S3 (feature columns); S5 (forecast methods and numbers); S3 and S4 (Tableau dashboard).
Hands over to	Everyone (project repo, report and slides).

What you must do

1. Set up the shared **GitHub repository** with the folders: data/raw, data/clean, notebooks, sql, dashboard and report.
2. Start the project documentation and collect the project requirements in one document.
3. Learn the ticket-class columns in the data (for example Standard and First).
4. Prepare daily aggregated tables per ticket class and train first test models together with S5.
5. Forecast the **demand for each ticket class** (Standard versus First) for next month.
6. Lead the report and slides: collect every student's section, keep one consistent style, and check that nothing is missing.
7. Organize the final presentation so that everyone presents their own part.

What you deliver

- GitHub repository with the agreed folders
- Requirements document
- Forecast plots for ticket class demand
- Final report and presentation slides

You are done when

The repository is organized, the class demand forecast is complete, and the report and slides contain every student's section.

Project deliverables and owners

Deliverable	Owner
Cleaned dataset + preprocessing notebook	S1 + S3, reviewed by S4
Data model diagram	S2
Document of analysis questions (each marked SQL or pandas, with a sample chart)	S3 (operations), S4 (revenue), S2 (SQL), S1 (pandas)
Forecast plots (rides, daily revenue, class demand)	S5, S6
Tableau dashboard	S3 + S4
Final report and presentation	S6 leads; everyone writes and presents their own section

5. The handoff flow (who finishes first)

Step	Who	Hands off to	What is handed over
1	S1	S2, S3, S4	First cleaned draft (clean_v1) + data dictionary
2	S2	Everyone	SQL database loaded with the clean data
3	S3	S4, S5, S6	Added feature columns (day, month, hour, delay minutes...)
4	S4	S1	Validation report: problems found in the cleaning
5	S1	Everyone	Final cleaned dataset (clean_v2 or later)
6	S3, S4, S2	S5, S6, Tableau	Analysis answers + SQL queries + exported tables
7	S5, S6	S3, S4, S6	Forecast plots and numbers
8	S3 + S4	S6	Tableau dashboard
9	S6	Everyone	Report + slides, then everyone presents

Order of finishing: S1 first (clean draft), then S2 (database) and S3 (features), then S4 (validation). After that, the work follows steps 5 to 9 in the table above.

6. Team rules (they prevent chaos)

- **One shared GitHub repo** with folders: data/raw, data/clean, notebooks, sql, dashboard, report.
- **Never edit the raw data.** Only S1 changes the cleaned file, and every version gets a number (clean_v1.csv, clean_v2.csv).
- **Same column names for everyone.** S1 writes a short data dictionary listing each column and its meaning.
- **Short meeting every 2-3 days:** each person answers: what I finished, what I'm stuck on, who I'm waiting for.
- **Everyone reviews someone else's work,** because you learn faster that way and catch mistakes the author cannot see.

7. Why does S4 test S1's cleaning?

The person who wrote the cleaning code has blind spots: they believe their own code works. A second person checks the result with fresh eyes against a written checklist, so mistakes are found before they spread into the SQL tables, the analysis and the forecasts.